

■ ■ Spidy Task Automation Assistant

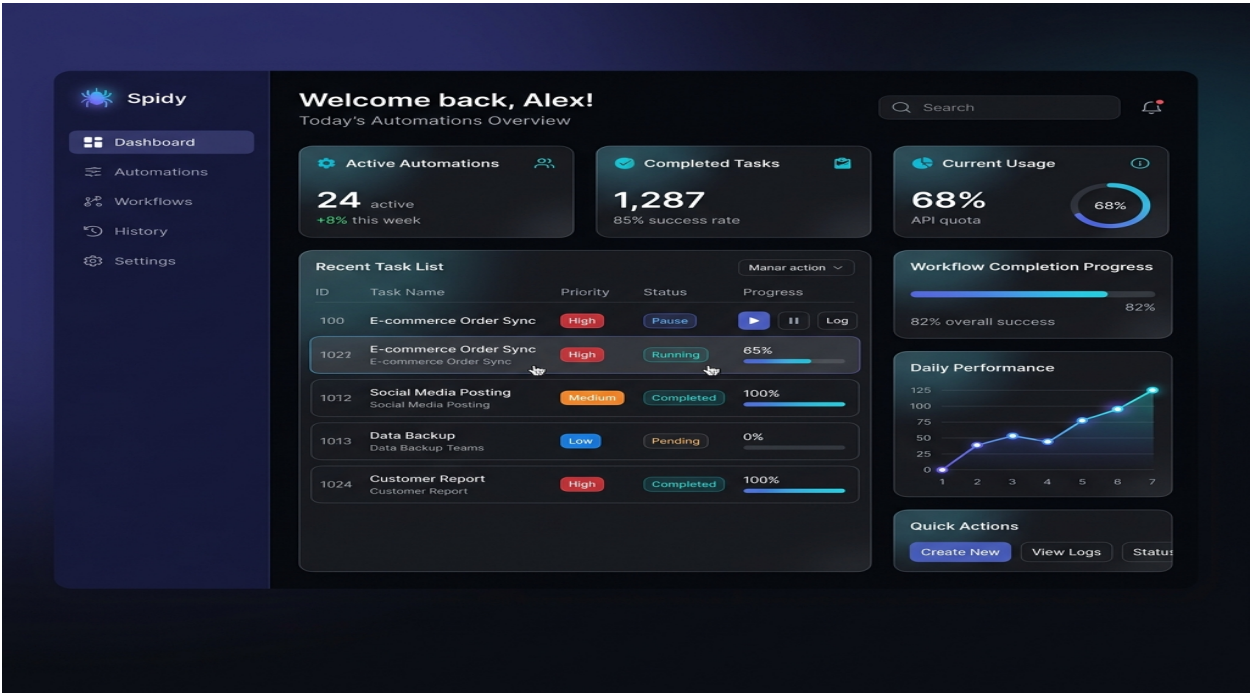
Official Interactive User Manual & Technical Specification

Executive Overview

Spidy Task Automation Assistant is a full-stack, AI-powered task management and automation platform built with **FastAPI** (Python), **SQLAlchemy ORM**, **SQLite**, and **React 18** (Vite + Tailwind CSS). It provides visual task management alongside natural language AI commands, specialized AI Agents, autonomous subtask solver workflows, and user credentials management.

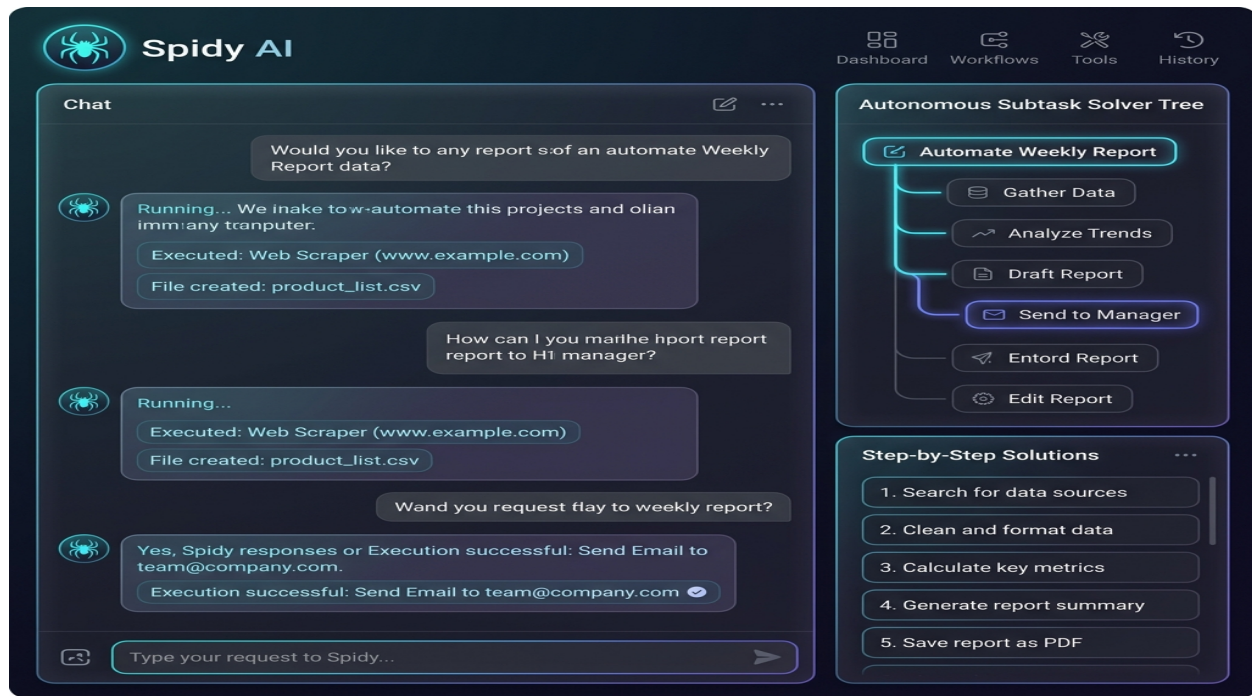
1. Interactive Visual Dashboard

The main dashboard provides real-time task progress metrics, urgency indicators, task search, category filters, and subtask accordion trees.



2. AI Natural Language Copilot Chat

The AI Copilot allows users to perform CRUD operations via natural language. Every function call generates a glowing **Tool Execution Badge** verifying function name, parsed JSON parameters, and execution safety status.



3. Specialized Free AI Agents

- **Wikipedia Encyclopedia Knowledge Agent (search_encyclopedia):** Queries free Wikipedia REST API to inject factual research summaries.
- **Open-Meteo Weather Agent (get_weather_info):** Queries free Open-Meteo API to return live weather forecasts.

4. Autonomous AI Subtask Solver & Tracking Engine

The Autonomous Subtask Solver takes a high-level goal, decomposes it into subtasks, invokes the Gemini LLM solver for every subtask, generates concrete resolution text/code blocks, and updates execution state (*pending* → *in_progress* → *completed*).

5. User Email Identity & Analytics Profile

Users can register/switch their Email ID and configure custom Google Gemini API Key credentials. The platform tracks task completion score percentages, total LLM operations, and subtasks solved by Gemini.



6. Quick Reference API Table

Action	Trigger / Route	HTTP Endpoint
Create Task	Header '+ New Task'	POST /api/tasks
AI Chat & Tool Execution	AI Assistant -> Chat	POST /api/ai/chat
Auto-Solve Goal	AI Assistant -> Auto-Solve	POST /api/ai/auto-execute-goal
User Login & Credentials	Header Email Badge	POST /api/users/login
User Analytics Profile	Settings -> User Analytics	GET /api/users/analytics